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Rong-Qian Linghu & Bing-Zhao Li

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Research Article



Quadratic-phase Laplace transform: definition and properties

Rong-Qian Linghu^a and Bing-Zhao Li^{a,b}

^aSchool of Mathematics and Statistics, Beijing Institute of Technology, Beijing, People's Republic of China;

^bBeijing Key Laboratory on MCAACI, Beijing Institute of Technology, Beijing, People's Republic of China

ABSTRACT

Eigenfunctions of a non-self-adjoint second-order differential operator T with complex coefficients are typically endowed with an intrinsic quadratic phase structure, which prevents classical Laplace and Fourier transforms from effectively characterizing its spectral properties. In this paper, a class of quadratic-phase Laplace transforms (QPLT) is constructed from differential operators with quadratic phase structures, whose kernel is shown to be the unique solution of the spectral problem associated with the operator T . Subsequently, the inversion formula and the Parseval identity for the QPLT are established, and its operational properties, including shifting, scaling, differentiation, convolution, and correlation theorems, are derived. Finally, the Donoho–Stark uncertainty principle for the QPLT is proved. The QPLT is demonstrated to provide an effective analytical tool for the study of non-self-adjoint quadratic operators and the associated partial differential equations.

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1. Introduction

In recent years, second order differential operators with complex coefficients defined on $\mathbb{R}^d$ have attracted widespread attention in spectral theory and evolution equation analysis. These operators are typically non-self-adjoint, and the combined effects of non-self-adjointness and quadratic terms present a series of fundamental challenges in spectral analysis and group evolution studies [1,2]. Specifically, the spectra of such operators are often distributed in the complex plane rather than along the real axis, and may exhibit sectorial or strip like structures, complicating spectral decomposition [3–5]. Moreover, these operators are often non-diagonalizable, accompanied by nontrivial Jordan structures, leading to transient growth and non-orthogonality effects in the evolution semigroup [3–5].

In response to these challenges, existing research has primarily focused on systematic analysis of operator spectral theory and semigroup evolution. Hitrik–Pravda-Starov revealed the intrinsic connection between the spectral structure of non-self-adjoint second-order operators and their singular space, thereby characterizing the smooth effects and convergence behaviour of the corresponding evolution semigroup, which forms the

CONTACT Bing-Zhao Li li_bingzhao@bit.edu.cn
Technology, Beijing 102488, People's Republic of China

School of Mathematics and Statistics, Beijing Institute of Technology, Beijing 102488, People's Republic of China

Beijing Key Laboratory on MCAACI, Beijing Institute of Technology, Beijing 102488, People's Republic of China

theoretical foundation for spectral analysis of complex quadratic-type operators [6]. Building on this foundation, Trapasso further introduced phase space wavepacket analysis and Gabor expansion methods, studying the propagation and decay properties of evolution semigroups generated by second-order differential operators with complex coefficients from the perspective of modulation spaces [7].

However, the eigenfunctions and generalized eigenfunctions of non-self-adjoint second-order operators T inherently exhibit irreducible higher-order phase structures, causing their solution representations to deviate from the purely exponential form. In particular, solutions to the spectral problem associated with T typically take the form of a product of an exponential decay factor and a quadratic phase term. This higher-order phase is intrinsically induced by the quadratic components of the operator and cannot be removed by classical integral transforms. Motivated by this observation, this paper considers a second-order differential operator T , then

$$T = \left(\frac{d^2}{dt^2} - 4iAt \frac{d}{dt} - 4iA^2 t^2 - 2iA \right), \quad A \in \mathbb{R}$$

with a quadratic phase structure and constructs a Laplace transform incorporating a quadratic phase, derived from the solution $u(t, \lambda) = e^{Ait^2 - \lambda t}$ representation of its spectral problem. Then, the quadratic-phase Laplace transform (QPLT) is defined as

$$\mathcal{Q}(f)[\lambda] = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(t) u(t, \lambda) dt, \quad \lambda = \sigma + \omega i.$$

Furthermore, its fundamental operational properties, convolution theorem, correlation theorem, and the Donoho–Stark-type uncertainty principle are thoroughly investigated [8,9].

The remainder of this paper is structured as follows: In Section 2, the QPLT is derived, and several basic properties of this transform are investigated. In Section 3, the paper is concluded.

2. Main results

2.1. Quadratic-phase Laplace operator

Definition 2.1: Let $u(t, \lambda) \in C^2(\mathbb{R})$, $\lambda \in \mathbb{C}$, and the parameter $A \in \mathbb{R}$. Then the quadratic-phase Laplace differential operator is defined as

$$Tu(t, \lambda) = \left(\frac{d^2}{dt^2} - 4iAt \frac{d}{dt} - 4iA^2 t^2 - 2iA \right) u(t, \lambda), \quad (2.1)$$

where $\lambda = \sigma + \omega i$, $\sigma \geq 0$, the T plays the role of the generating operator associated with the kernel.

Based on the differential operator defined above, the solution structure of its corresponding spectral problem can induce the construction form of the transform kernel, thereby establishing the following theorem for solving the QPLT kernel.

Theorem 2.1: Let $u(t, \lambda) \in C^2(\mathbb{R})$, $\lambda = \sigma + \omega i$, $\sigma \geq 0$, and $A \in \mathbb{R}$. Then the differential equation

$$\begin{cases} Tu(t, \lambda) = \lambda^2 u(t, \lambda), \\ u(0, \lambda) = 1, \quad u_t(0, \lambda) = -\lambda, \end{cases} \quad (2.2)$$

has a unique solution $u(t, \lambda)$, called the quadratic-phase Laplace kernel, we have $u(t, \lambda) = e^{Ait^2 - \lambda t}$.

Proof: By Definition 2.1 and Equation (2.2), expanding the operator T yields the following ordinary differential equation:

$$u'' - 4iAt u' - (4A^2 t^2 + 2iA + \lambda^2)u = 0.$$

This is a second-order linear ordinary differential equation. Writing it in the standard form

$$u'' + P(t)u' + Q(t)u = 0,$$

we have

$$P(t) = -4iAt \quad \text{and} \quad Q(t) = -(4A^2 t^2 + 2iA + \lambda^2).$$

Next, let $u = \phi v$, where ϕ is a nonvanishing twice differentiable function. We obtain

$$u'(t) = \phi'(t)v(t) + \phi(t)v'(t).$$

Differentiating once more gives

$$u''(t) = \phi''(t)v(t) + 2\phi'(t)v'(t) + \phi(t)v''(t).$$

Substituting these expressions into the differential equation and dividing by $\phi(t)$, we obtain

$$v'' + \left(2\frac{\phi'}{\phi} + P(t)\right)v' + \left(\frac{\phi''}{\phi} + P(t)\frac{\phi'}{\phi} + Q(t)\right)v = 0.$$

According to the Liouville transformation, ϕ is chosen such that the coefficient of v' vanishes. Thus,

$$2\frac{\phi'}{\phi} + P(t) = 0.$$

Since $P(t) = -4iAt$, it follows that

$$\frac{\phi'}{\phi} = -\frac{P(t)}{2} = 2iAt.$$

Integrating this equation gives

$$\phi(t) = C e^{iAt^2}.$$

Where C is a nonzero constant. Taking the normalization $\phi(0) = 1$, we obtain $C = 1$ and hence

$$\phi(t) = e^{iAt^2}.$$

Moreover,

$$\frac{\phi'}{\phi} = 2iAt \quad \text{and} \quad \frac{\phi''}{\phi} = 2iA - 4A^2t^2.$$

Therefore, we obtain

$$\begin{aligned} \frac{\phi''}{\phi} + P(t)\frac{\phi'}{\phi} + Q(t) &= (2iA - 4A^2t^2) + (-4iAt)(2iAt) \\ &\quad - (4A^2t^2 + 2iA + \lambda^2) \\ &= -\lambda^2. \end{aligned}$$

Consequently, the differential equation for v reduces to

$$v'' - \lambda^2 v = 0.$$

Since $\phi(0) = 1$ and $\phi'(0) = 0$, the initial conditions

$$u(0) = 1 \quad \text{and} \quad u'(0) = -\lambda$$

imply

$$v(0) = 1 \quad \text{and} \quad v'(0) = -\lambda.$$

Thus, the resulting initial-value problem is

$$\begin{cases} v''(t) = \lambda^2 v(t), \\ v(0) = 1, \\ v'(0) = -\lambda. \end{cases}$$

For $\lambda \neq 0$, the general solution is

$$v(t) = c_1 e^{\lambda t} + c_2 e^{-\lambda t}.$$

Applying the initial conditions gives

$$c_1 + c_2 = 1 \quad \text{and} \quad \lambda(c_1 - c_2) = -\lambda.$$

Hence, $c_1 = 0$ and $c_2 = 1$, so that

$$v(t) = e^{-\lambda t}.$$

When $\lambda = 0$, the equation becomes $v'' = 0$. The initial conditions $v(0) = 1$ and $v'(0) = 0$ give $v(t) = 1$, which is also consistent with $v(t) = e^{-\lambda t}$. Therefore, for every $\lambda \in \mathbb{C}$,

$$v(t) = e^{-\lambda t}.$$

Finally, substituting $\phi(t) = e^{iAt^2}$ and $v(t) = e^{-\lambda t}$ into $u = \phi v$, we obtain the quadratic-phase Laplace kernel

$$u(t, \lambda) = e^{iAt^2 - \lambda t}.$$

This completes the proof. ■

After the explicit expression of the quadratic-phase Laplace kernel is established, the QPLT can be further defined based on this kernel function.

2.2. quadratic-phase Laplace transform

Definition 2.2: Let $f(t) \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, $\lambda \in \mathbb{C}$ and $A \in \mathbb{R}$. Then the QPLT is defined as

$$\mathcal{Q}(f)[\lambda] = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(t) u(t, \lambda) dt, \quad \lambda = \sigma + \omega i, \quad (2.3)$$

where the $f(t)$ satisfying $\int_{\mathbb{R}} |f(t)| e^{-\sigma t} dt < \infty$, $\sigma \geq 0$.

Based on the above definition of the transform, its fundamental properties will be systematically studied next, including the inversion formula as well as several important operational properties.

Theorem 2.2: (Inversion formula) $\mathcal{Q}(f)[\sigma + i\omega]$ is analytic in the half-plane $\text{Re}(\lambda) > \sigma_0$, where $\lambda = \sigma + i\omega$ and $\text{Re}(\lambda)$ denotes the real part of λ , so that $\text{Re}(\lambda) = \sigma$. For any fixed $\sigma > \sigma_0$. Then the inversion formula is given by

$$f(t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \mathcal{Q}(f)[\sigma + \omega i] e^{(\sigma + \omega i)t - iAt^2} d\omega, \quad (2.4)$$

where $\mathcal{Q}(f)[\sigma + \omega i]$ is the QPLT for $f(t)$.

Proof: Let the $g(t) = f(t) e^{iAt^2}$, then

$$\mathcal{Q}(f)[\lambda] = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(t) e^{iAt^2} e^{-\lambda t} dt = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} g(t) e^{-\lambda t} dt.$$

According to $\lambda = \sigma + \omega i$, then $d\lambda = i d\omega$, we obtain

$$\begin{aligned} g(t) &= \frac{\sqrt{2\pi}}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} \mathcal{Q}(f)[\lambda] e^{\lambda t} d\lambda \\ &= \frac{\sqrt{2\pi}}{2\pi i} i \int_{\mathbb{R}} \mathcal{Q}(f)[\sigma + \omega i] e^{(\sigma + \omega i)t} d\omega \\ &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \mathcal{Q}(f)[\sigma + \omega i] e^{(\sigma + \omega i)t} d\omega. \end{aligned}$$

Finally, we obtain

$$f(t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \mathcal{Q}(f)[\sigma + \omega i] e^{(\sigma + \omega i)t - iAt^2} d\omega.$$

This completes the proof. ■

Theorem 2.3: (Parseval formula) Let $f, g \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, let $A \in \mathbb{R}$, and let $\sigma \geq 0$ be fixed. Assume that $e^{-\sigma(\cdot)} f$ and $e^{-\sigma(\cdot)} g \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$. Then the Parseval formula is given

by

$$\int_{\mathbb{R}} f(t) \overline{g(t)} e^{-2\sigma t} dt = \int_{\mathbb{R}} \mathcal{Q}(f)[\sigma + i\omega] \overline{\mathcal{Q}(g)[\sigma + i\omega]} d\omega. \quad (2.5)$$

In particular, by taking $g = f$, we obtain

$$\int_{\mathbb{R}} |f(t)|^2 e^{-2\sigma t} dt = \int_{\mathbb{R}} |\mathcal{Q}(f)[\sigma + i\omega]|^2 d\omega.$$

Equivalently,

$$\|e^{-\sigma(\cdot)} f\|_{L^2(\mathbb{R})} = \|\mathcal{Q}(f)[\sigma + i(\cdot)]\|_{L^2(\mathbb{R})}.$$

When $\sigma = 0$, the above identity reduces to

$$\|f\|_{L^2(\mathbb{R})} = \|\mathcal{Q}(f)[i(\cdot)]\|_{L^2(\mathbb{R})}.$$

Proof: Define

$$F_{\sigma}(t) = f(t) e^{iAt^2 - \sigma t} \quad \text{and} \quad G_{\sigma}(t) = g(t) e^{iAt^2 - \sigma t}.$$

By Definition 2.2, for $\lambda = \sigma + i\omega$, we have

$$\begin{aligned} \mathcal{Q}(f)[\sigma + i\omega] &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(t) e^{iAt^2 - \sigma t - i\omega t} dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} F_{\sigma}(t) e^{-i\omega t} dt = \mathcal{F}(F_{\sigma})[\omega]. \end{aligned}$$

Similarly, we obtain

$$\mathcal{Q}(g)[\sigma + i\omega] = \mathcal{F}(G_{\sigma})[\omega],$$

where $\mathcal{F}$ denotes the Fourier transform with the normalization factor $1/\sqrt{2\pi}$. Since $F_{\sigma}, G_{\sigma} \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, the Parseval identity for the Fourier transform yields

$$\begin{aligned} &\int_{\mathbb{R}} \mathcal{Q}(f)[\sigma + i\omega] \overline{\mathcal{Q}(g)[\sigma + i\omega]} d\omega \\ &= \int_{\mathbb{R}} F_{\sigma}(t) \overline{G_{\sigma}(t)} dt. \end{aligned}$$

Because A , σ , and t are real, we have

$$\begin{aligned} F_{\sigma}(t) \overline{G_{\sigma}(t)} &= f(t) e^{iAt^2 - \sigma t} \overline{g(t) e^{iAt^2 - \sigma t}} \\ &= f(t) e^{iAt^2 - \sigma t} \overline{g(t)} e^{-iAt^2 + \sigma t} \\ &= f(t) \overline{g(t)} e^{-2\sigma t}. \end{aligned}$$

Therefore,

$$\int_{\mathbb{R}} f(t) \overline{g(t)} e^{-2\sigma t} dt = \int_{\mathbb{R}} \mathcal{Q}(f)[\sigma + i\omega] \overline{\mathcal{Q}(g)[\sigma + i\omega]} d\omega.$$

Taking $g = f$ gives

$$\int_{\mathbb{R}} |f(t)|^2 e^{-2\sigma t} dt = \int_{\mathbb{R}} |\mathcal{Q}(f)[\sigma + i\omega]|^2 d\omega.$$

Finally, when $\sigma = 0$, we obtain

$$\|f\|_{L^2(\mathbb{R})} = \|\mathcal{Q}(f)[i(\cdot)]\|_{L^2(\mathbb{R})}.$$

This completes the proof. ■

Property 2.1: (Linearity) Let $f(t), g(t) \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, $\alpha, \beta \in \mathbb{R}$. Then the linearity is given by

$$\mathcal{Q}(\alpha f + \beta g)[\lambda] = \alpha \mathcal{Q}(f)[\lambda] + \beta \mathcal{Q}(g)[\lambda]. \quad (2.6)$$

Proof: By Definition 2.2, we obtain

$$\begin{aligned} \mathcal{Q}(\alpha f + \beta g)[\lambda] &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} (\alpha f(t) + \beta g(t)) e^{Ait^2 - \lambda t} dt \\ &= \frac{1}{\sqrt{2\pi}} \left(\alpha \int_{\mathbb{R}} f(t) e^{Ait^2 - \lambda t} dt + \beta \int_{\mathbb{R}} g(t) e^{Ait^2 - \lambda t} dt \right) \\ &= \alpha \mathcal{Q}(f)[\lambda] + \beta \mathcal{Q}(g)[\lambda]. \end{aligned}$$

This completes the proof. ■

Property 2.2: (Time shift) Let $f(t) \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, $f_{t_0}(t) = f(t - t_0)$ and $t_0 \in \mathbb{R}$. Then

$$\mathcal{Q}(f_{t_0})[\lambda] = e^{Ait_0^2 - \lambda t_0} \mathcal{Q}(f)[\lambda - 2Ait_0]. \quad (2.7)$$

Proof: By Definition 2.2, let the $u = t - t_0$, we obtain

$$\begin{aligned} \mathcal{Q}(f_{t_0})[\lambda] &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(u) e^{Ai(u+t_0)^2 - \lambda(u+t_0)} du \\ &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(u) e^{Ai(u^2 + 2ut_0 + t_0^2) - \lambda u - \lambda t_0} du \\ &= e^{Ait_0^2 - \lambda t_0} \cdot \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(u) e^{Aiu^2 - (\lambda - 2Ait_0)u} du \\ &= e^{Ait_0^2 - \lambda t_0} \mathcal{Q}(f)[\lambda - 2Ait_0]. \end{aligned}$$

This completes the proof. ■

Property 2.3: (Frequency shift) Let $f(t) \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, $g(t) = e^{iw_0 t} f(t)$, and $w_0 \in \mathbb{R}$. Then the frequency shift is given by

$$\mathcal{Q}(g)[\lambda] = \mathcal{Q}_A(f)[\lambda - iw_0] \quad (2.8)$$

Proof: By Definition 2.2, we obtain

$$\begin{aligned} \mathcal{Q}(g)[\lambda] &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{iw_0 t} f(t) e^{Ait^2 - \lambda t} dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(t) e^{Ait^2 - (\lambda - iw_0)t} dt \\ &= \mathcal{Q}(f)[\lambda - iw_0]. \end{aligned}$$

This completes the proof. ■

Property 2.4: (Scaling) Let $f(t) \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, $f_m(t) = f(mt)$ and $m > 0$. Then the scaling is given by

$$\mathcal{Q}_A(f_m)[\lambda] = \frac{1}{m} \mathcal{Q}_{A/m^2}(f) \left[\frac{\lambda}{m} \right]. \quad (2.9)$$

Proof: By Definition 2.2, let the $x = mt$, we obtain

$$\begin{aligned} \mathcal{Q}_A(f_m)[\lambda] &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(mt) e^{Ait^2 - \lambda t} dt \\ &= \frac{1}{m\sqrt{2\pi}} \int_{\mathbb{R}} f(x) e^{i(A/m^2)x^2 - (\lambda/m)x} dx \\ &= \frac{1}{m} \mathcal{Q}_{A/m^2}(f) \left[\frac{\lambda}{m} \right], \end{aligned}$$

This completes the proof. ■

Property 2.5: (Differentiation) Let $f(t) \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, $f \in C^\infty(\mathbb{R})$. Then the differentiation is given by

$$\frac{\partial^n}{\partial \lambda^n} \mathcal{Q}(f)[\lambda] = (-1)^n \mathcal{Q}(t^n f(t))[\lambda], \quad (2.10)$$

and

$$\mathcal{Q}(f^{(n)})[\lambda] = \left(\lambda + 2Ai \frac{\partial}{\partial \lambda} \right)^n \mathcal{Q}(f)[\lambda]. \quad (2.11)$$

Proof: Differentiation with respect to λ , we obtain

$$\frac{\partial}{\partial \lambda} \mathcal{Q}(f)[\lambda] = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(t) \frac{\partial}{\partial \lambda} (e^{Ait^2 - \lambda t}) dt.$$

Next, we obtain

$$\frac{\partial}{\partial \lambda} (e^{Ait^2 - \lambda t}) = -te^{Ait^2 - \lambda t}.$$

Substituting into the original transform, we obtain,

$$\frac{\partial}{\partial \lambda} \mathcal{Q}_A(f)[\lambda] = -\mathcal{Q}_A(tf)[\lambda].$$

Equation (2.10) follows directly by induction. Next, we prove the differentiation with respect to t , and let the boundary term vanish

$$\lim_{t \rightarrow \pm\infty} f(t)e^{Ait^2 - \lambda t} = 0.$$

Next, we have

$$\begin{aligned} \mathcal{Q}(f')[\lambda] &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f'(t)e^{Ait^2 - \lambda t} dt \\ &= \frac{1}{\sqrt{2\pi}} [f(t)e^{Ait^2 - \lambda t}]_{-\infty}^{\infty} - \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(t) \frac{d}{dt} (e^{Ait^2 - \lambda t}) dt \\ &= -\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(t)(2Ait - \lambda)e^{Ait^2 - \lambda t} dt \\ &= \lambda \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(t)e^{Ait^2 - \lambda t} dt - 2Ai \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} [tf(t)]e^{Ait^2 - \lambda t} dt \\ &= \lambda \mathcal{Q}(f)[\lambda] - 2Ai \mathcal{Q}(tf)[\lambda] \\ &= \left(\lambda + 2Ai \frac{\partial}{\partial \lambda} \right) \mathcal{Q}(f)[\lambda]. \end{aligned}$$

Obviously, the Equation (2.11) follows directly by induction.

This completes the proof. ■

2.3. Convolution theorem and correlation theorem

The convolution theorem reveals the structural characteristics of the transform under convolution operations, laying the theoretical foundation for frequency domain analysis and efficient computation of complex systems. The correlation theorem establishes a mathematical correspondence between the time frequency relationships of signals and the properties in the transform domain, playing a critical role in fields such as signal detection and pattern recognition. Many scholars are dedicated to deepening the theoretical framework of these theorems and expanding their interdisciplinary applications, continuously advancing the development of this field [4,10–13].

Definition 2.3: Let $f(t), g(t) \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ and $A, \tau \in \mathbb{R}$, the convolution operators $(f \star_A g)(t)$ and correlation operators $(f \star_A g)(t)$ for the QPLT are respectively defined as

$$(f \star_A g)(t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(\tau)g(t - \tau)e^{-2Ai\tau(t-\tau)} d\tau, \quad (2.12)$$

and

$$(f \star_A g)(t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(\tau)g(\tau + t)e^{2Ai\tau(\tau+t)} d\tau. \quad (2.13)$$

Theorem 2.4: Let $f(t), g(t) \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, $\lambda \in \mathbb{C}$ and $A \in \mathbb{R}$, the convolution theorem $\mathcal{Q}(f \star_A g)[\lambda]$ and correlation theorem $\mathcal{Q}(f \star_A g)[\lambda]$ for the QPLT are respectively defined as

$$\mathcal{Q}(f \star_A g)[\lambda] = \mathcal{Q}(f)[\lambda]\mathcal{Q}(g)[\lambda], \quad (2.14)$$

and

$$\mathcal{Q}(f \star_A g)[\lambda] = \mathcal{Q}(f)[- \lambda]\mathcal{Q}(g)[\lambda]. \quad (2.15)$$

Proof: Convolution theorem: by Definition 2.2 and Equation (2.12), we have

$$\begin{aligned} \mathcal{Q}(f \star_A g)[\lambda] &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} (f \star_A g)(t)e^{Ait^2 - \lambda t} dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \left(\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(\tau)g(t - \tau)e^{-2Ai\tau(t-\tau)} d\tau \right) e^{Ait^2 - \lambda t} dt \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} \int_{\mathbb{R}} f(\tau)g(t - \tau)e^{-2Ai\tau(t-\tau)} e^{Ait^2 - \lambda t} d\tau dt. \end{aligned}$$

Let the $s = t - \tau$, we obtain

$$\mathcal{Q}(f \star_A g)[\lambda] = \frac{1}{2\pi} \int_{\mathbb{R}} \int_{\mathbb{R}} f(\tau)g(s)e^{-2Ai\tau s} e^{Ai(\tau+s)^2 - \lambda(\tau+s)} d\tau ds.$$

The exponential part simplifies to

$$e^{-2Ai\tau s} e^{Ai(\tau+s)^2 - \lambda(\tau+s)} = e^{Ai\tau^2 - \lambda\tau} e^{Ais^2 - \lambda s}.$$

Next, we obtain

$$\begin{aligned} \mathcal{Q}(f \star_A g)[\lambda] &= \frac{1}{2\pi} \int_{\mathbb{R}} \int_{\mathbb{R}} f(\tau)g(s)e^{Ai\tau^2 - \lambda\tau} e^{Ais^2 - \lambda s} d\tau ds \\ &= \left(\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(\tau)e^{Ai\tau^2 - \lambda\tau} d\tau \right) \left(\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} g(s)e^{Ais^2 - \lambda s} ds \right) \\ &= \mathcal{Q}(f)[\lambda]\mathcal{Q}(g)[\lambda]. \end{aligned}$$

The convolution theorem has been proved. Next, we prove the correlation theorem by using Definition 2.2 and Equation (2.13), we have

$$\mathcal{Q}(f \star_A g)[\lambda] = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} (f \star_A g)(t)e^{Ait^2 - \lambda t} dt$$

$$\begin{aligned}
 &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \left(\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(\tau) g(t + \tau) e^{2Ai\tau(t+\tau)} d\tau \right) e^{Ai t^2 - \lambda t} dt \\
 &= \frac{1}{2\pi} \int_{\mathbb{R}} \int_{\mathbb{R}} f(\tau) g(t + \tau) e^{2Ai\tau(t+\tau)} e^{Ai t^2 - \lambda t} d\tau dt.
 \end{aligned}$$

Let the $s = t + \tau$, we obtain

$$\mathcal{Q}(f \star_A g)[\lambda] = \frac{1}{2\pi} \int_{\mathbb{R}} \int_{\mathbb{R}} f(\tau) g(s) e^{2Ai\tau s} e^{Ai(s-\tau)^2 - \lambda(s-\tau)} d\tau ds.$$

The exponential part simplifies to

$$e^{2Ai\tau s} e^{Ai(s-\tau)^2 - \lambda(s-\tau)} = e^{Ai\tau^2 + \lambda\tau} e^{Ais^2 - \lambda s}.$$

Next, we obtain

$$\begin{aligned}
 \mathcal{Q}(f \star_A g)[\lambda] &= \frac{1}{2\pi} \int_{\mathbb{R}} \int_{\mathbb{R}} f(\tau) g(s) e^{Ai\tau^2 + \lambda\tau} e^{Ais^2 - \lambda s} d\tau ds \\
 &= \left(\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(\tau) e^{Ai\tau^2 + \lambda\tau} d\tau \right) \left(\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} g(s) e^{Ais^2 - \lambda s} ds \right) \\
 &= \mathcal{Q}(f)[- \lambda] \mathcal{Q}(g)[\lambda].
 \end{aligned}$$

This completes the proof. ■

2.4. Uncertainty principles

This section focuses on studying the Donoho–Stark-type uncertainty principle for the QPLT. This principle quantifies the constraints on signal energy distribution between the time domain and the transform domain, establishing rigorous mathematical bounds on signal localization. It holds fundamental theoretical significance in fields such as non-stationary signal processing and coherence analysis of quantum systems [14–18].

Theorem 2.5: Let $f(t) \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, $\lambda = \sigma + i\omega \in \mathbb{C}$ with $\sigma \geq 0$, and $A \in \mathbb{R}$. Let $\Omega, \Sigma \subset \mathbb{R}$ be measurable sets with $0 < |\Omega| < \infty$ and $0 < |\Sigma| < \infty$. Assume that $f(t)$ satisfies $\int_{\mathbb{R}} |f(t)| e^{-\sigma t} dt < \infty$, $\int_{\mathbb{R}} |f(t)|^2 e^{-2\sigma t} dt < \infty$, and that there exist constants $0 < \varepsilon_{\Omega} < 1$, $0 < \varepsilon_{\Sigma} < 1$ such that

$$\int_{\mathbb{R} \setminus \Omega} |f(t)| e^{-\sigma t} dt \leq \varepsilon_{\Omega} \int_{\mathbb{R}} |f(t)| e^{-\sigma t} dt, \quad (2.16)$$

and

$$\int_{\mathbb{R} \setminus \Sigma} |\mathcal{Q}(f)[\sigma + i\omega]|^2 d\omega \leq \varepsilon_{\Sigma}^2 \int_{\mathbb{R}} |f(t)|^2 e^{-2\sigma t} dt, \quad (2.17)$$

then the Donoho–Stark uncertainty principle associated with the QPLT is given by

$$(1 - \varepsilon_{\Omega})^2 (1 - \varepsilon_{\Sigma}^2) \leq \frac{|\Omega| |\Sigma|}{2\pi}. \quad (2.18)$$

Proof: Fix $\sigma \geq 0$ and define

$$g_\sigma(t) = f(t)e^{-\sigma t}e^{Ait^2}.$$

Let $\widehat{g}_\sigma$ denote the Fourier transform of g_σ , defined as

$$\widehat{g}_\sigma(\omega) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} g_\sigma(t)e^{-i\omega t} dt.$$

Then, by Definition 2.2, we obtain,

$$\mathcal{Q}(f)(\sigma + i\omega) = \widehat{g}_\sigma(\omega).$$

Since $|e^{Ait^2}| = 1$, we obtain

$$\|g_\sigma\|_1 = \int_{\mathbb{R}} |f(t)|e^{-\sigma t} dt, \quad \|g_\sigma\|_2^2 = \int_{\mathbb{R}} |f(t)|^2 e^{-2\sigma t} dt.$$

According to the Plancherel formula for the Fourier transform, we obtain $\|\widehat{g}_\sigma\|_2 = \|g_\sigma\|_2$. Next, according to the Equation (2.16), we obtain

$$(1 - \varepsilon_\Omega) \int_{\mathbb{R}} |g_\sigma(t)| dt = (1 - \varepsilon_\Omega) \|g_\sigma\|_1 \leq \int_{\Omega} |g_\sigma(t)| dt, \quad (2.19)$$

by the Cauchy-Schwarz, we obtain

$$\left(\int_{\Omega} |g_\sigma(t)| dt \right)^2 \leq \left(\int_{\Omega} 1^2 dt \right) \left(\int_{\Omega} |g_\sigma(t)|^2 dt \right) = |\Omega| \int_{\Omega} |g_\sigma(t)|^2 dt \leq |\Omega| \|g_\sigma\|_2^2, \quad (2.20)$$

combining the Equations (2.19) and (2.20), we obtain

$$\frac{(1 - \varepsilon_\Omega)^2}{|\Omega|} \|g_\sigma\|_1^2 \leq \|g_\sigma\|_2^2. \quad (2.21)$$

According to the Plancherel formula for the Fourier transform and Equation (2.17), we obtain

$$\int_{\mathbb{R} \setminus \Sigma} |\widehat{g}_\sigma(\omega)|^2 d\omega \leq \varepsilon_\Sigma^2 \|\widehat{g}_\sigma\|_2^2 = \varepsilon_\Sigma^2 \|g_\sigma\|_2^2,$$

and

$$\int_{\Sigma} |\widehat{g}_\sigma(\omega)|^2 d\omega = \|\widehat{g}_\sigma\|_2^2 - \int_{\mathbb{R} \setminus \Sigma} |\widehat{g}_\sigma(\omega)|^2 d\omega \geq (1 - \varepsilon_\Sigma^2) \|g_\sigma\|_2^2. \quad (2.22)$$

Next, according to the Fourier transform,

$$|\widehat{g}_\sigma(\omega)| = \left| \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} g_\sigma(t)e^{-i\omega t} dt \right| \leq \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} |g_\sigma(t)| dt = \frac{1}{\sqrt{2\pi}} \|g_\sigma\|_1.$$

Since $\|\widehat{g}_\sigma\|_\infty \leq \frac{1}{\sqrt{2\pi}} \|g_\sigma\|_1$, we obtain

$$\int_{\Sigma} |\widehat{g}_\sigma(\omega)|^2 d\omega \leq |\Sigma| \|\widehat{g}_\sigma\|_\infty^2 \leq \frac{|\Sigma|}{2\pi} \|g_\sigma\|_1^2. \quad (2.23)$$

According to the Equations (2.22) and (2.23), we obtain

$$(1 - \varepsilon_{\Sigma}^2) \|g_{\sigma}\|_2^2 \leq \frac{|\Sigma|}{2\pi} \|g_{\sigma}\|_1^2. \quad (2.24)$$

Using the Equations (2.21) and (2.24), we obtain

$$(1 - \varepsilon_{\Sigma}^2) \frac{(1 - \varepsilon_{\Omega})^2}{|\Omega|} \|g_{\sigma}\|_1^2 \leq \frac{|\Sigma|}{2\pi} \|g_{\sigma}\|_1^2.$$

Finally, we obtain

$$(1 - \varepsilon_{\Omega})^2 (1 - \varepsilon_{\Sigma}^2) \leq \frac{|\Omega| |\Sigma|}{2\pi}.$$

This completes the proof. ■

3. Conclusion

This paper focuses on non-self-adjoint second order differential operators with quadratic phase structures and constructs a corresponding the QPLT. The transform's theoretical framework is established, including the inversion formula, Parseval identity, and theories related to convolution and correlation operations. Additionally, mathematical properties such as translation, scaling, and differentiation are rigorously proven. Through the analysis of the Donoho–Stark-type uncertainty principle, the theoretical characteristics of the transform in time frequency localization are revealed, providing an effective mathematical tool for the analysis of complex-coefficient quadratic-type operators.

Author contributions

CRedit: **Rong-Qian Linghu**: Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft; **Bing-Zhao Li**: Conceptualization, Funding acquisition, Investigation, Supervision, Writing – review & editing

Data availability statement

No data was used for the research described in this paper.

Disclosure statement

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